

TEMPEST: Soil Gas Well

November 2025 Samples

2026-09-24

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##Run Information

```
cat("Run Information: Brenden Blakley ") #lets you know what section you're in
```

Run Information: Brenden Blakley

```
#set the run date & user name
```

```
sample_year <- "2025"
```

```
sample_month <- "November"
```

```
user <- "Brenden Blakley"
```

```
#identify the files you want to read in
```

```
#read in as a list to accommodate multiple runs in a month (.txt file(s))
```

```
GHG_files <- c("Raw Data/20260827_TMP2025_Nov_Sept.txt")
```

```
#Metadata file path
```

```
Metadata_path = "Metadata/TEMPEST_SGAS_202511_ALL_WORKING.csv"
```

```
# Define the file path for QAQC log file - NO Need to change just check year
```

```
log_path <- "GC_QAQC_Log.csv"
```

```
check_log_path <- "GC_Check_QAQC_Log.csv"
```

```
# Define final path for data be sure to change year and month and run number
```

```
final_path <- "Processed Data/TMP2025_November.csv"
```

```
##Standard Type ("Actual" or "Half")
```

```
#LTREB, SMARTX, and GENX PW Samples use "Half", others should use "Actual"
```

```
Standard_Type <- "Actual"
```

```
#Did Some Samples require manual area retrieval?
```

```
##List Samples and their areas for each analyte ("CH4", "CO2", or "N2O")
```

```
Manual_Areas <- data.frame(
```

```
  Sample = c("NA"),
```

```
  Area = c("NA"),
```

```
  Analyte = c("NA")
```

```
)
```

```
#Gas in Water Calculations
```

```
#Are All Samples Gas in Water? ("YES" or "NO")
```

```
GIW <- "NO"
```

```
#Are some samples Gas in Water?
```

```
##List Sample IDs
```

```
GIW_Samples <- c()
```

```
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
```

```
TempC = 25 #Equilibration Temperature (Celsius)
```

```
volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrated
```

```
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrated (should be equal to volume_g)
```

```
#record any notes about the run or anything other info here:
```

```

run_notes <- ""

#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

cat(run_notes)

##Set Up
##Read in metadata and create similar sample IDs for matching to samples
##Read In Data Files
#Get Location of Each Sample in Main Data Frame
#Turn Each Sample into it's own Data Frame
#Get Only Area Tables and Retrieve Info
##QAQC Peak Count and Retention Time

## [1] "If there are too many peaks there is likely a problem with the sample or run and the chromatogr

## [1] "FID: Blank1; 1" "FID: Blank2; 1"
## [3] "FID: Oppm; 1" "FID: Std_10K_3mL; 1"
## [5] "FID: Std_10K_6mL; 1" "FID: Std_10K_12mL; 1"
## [7] "FID: Std_10K_24mL; 1" "FID: Blank3; 1"
## [9] "FID: Blank4; 1" "FID: Blank5; 1"
## [11] "FID: Blank6; 1" "FID: Blank7; 1"
## [13] "FID: ChkStd_HighTri_12mL_5; 3" "FID: Blank8; 1"
## [15] "FID: Blank9; 1" "FID: Blank10; 1"
## [17] "FID: Blank11; 1"

## [1] "ECD: Blank1; 0" "ECD: Blank2; 0" "ECD: Oppm; 0"
## [4] "ECD: Std_10K_3mL; 0" "ECD: Std_10K_6mL; 0" "ECD: Std_10K_12mL; 0"
## [7] "ECD: Std_10K_24mL; 0" "ECD: Blank3; 0" "ECD: Blank4; 0"
## [10] "ECD: Blank5; 0" "ECD: Blank6; 0" "ECD: Blank7; 0"
## [13] "ECD: Blank8; 0" "ECD: Blank9; 0" "ECD: Blank10; 0"
## [16] "ECD: Blank11; 0"

## [1] "If RTime is wrong check that correct number of peaks were found, if not a peak may have been mi

## [1] "CH4: "

## [1] "CO2: "

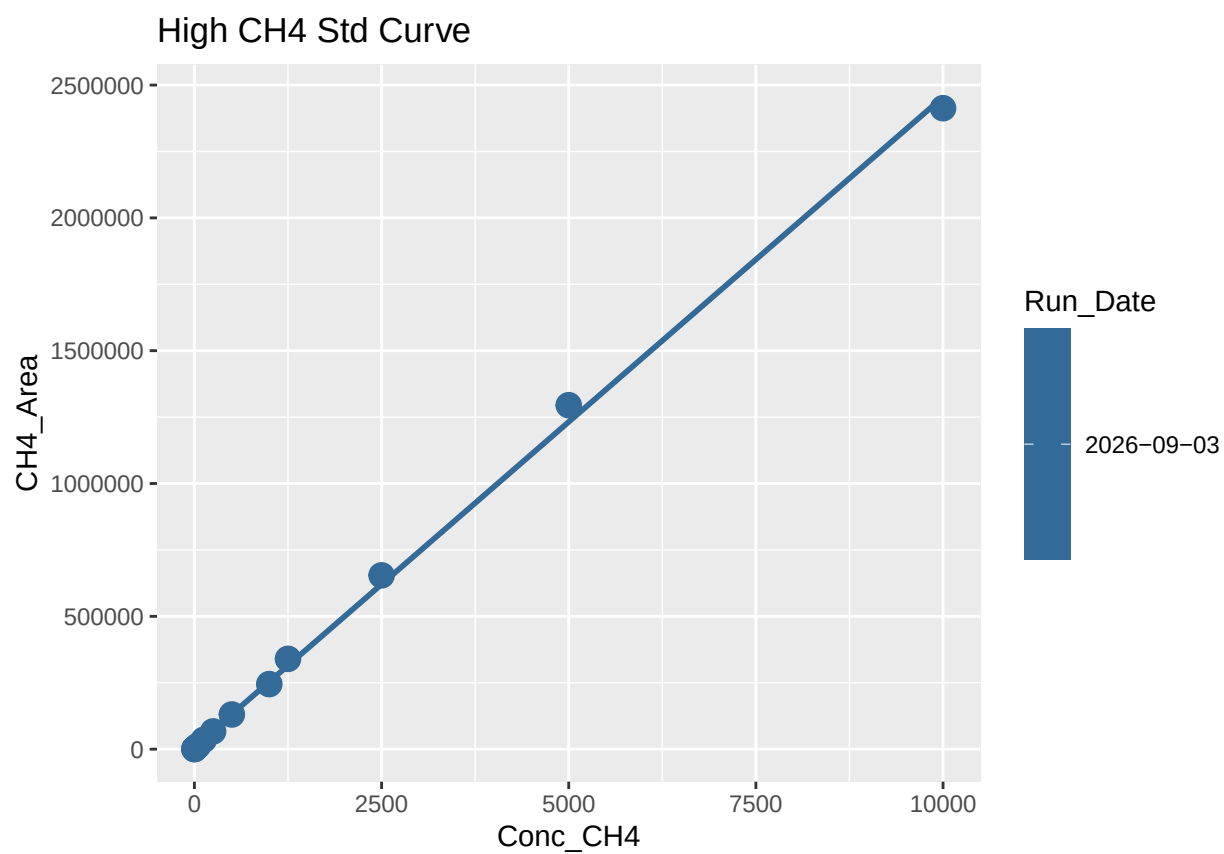
## [1] "N2O: "

##Format Data
##Add Standard Concentrations
##Finalize Data

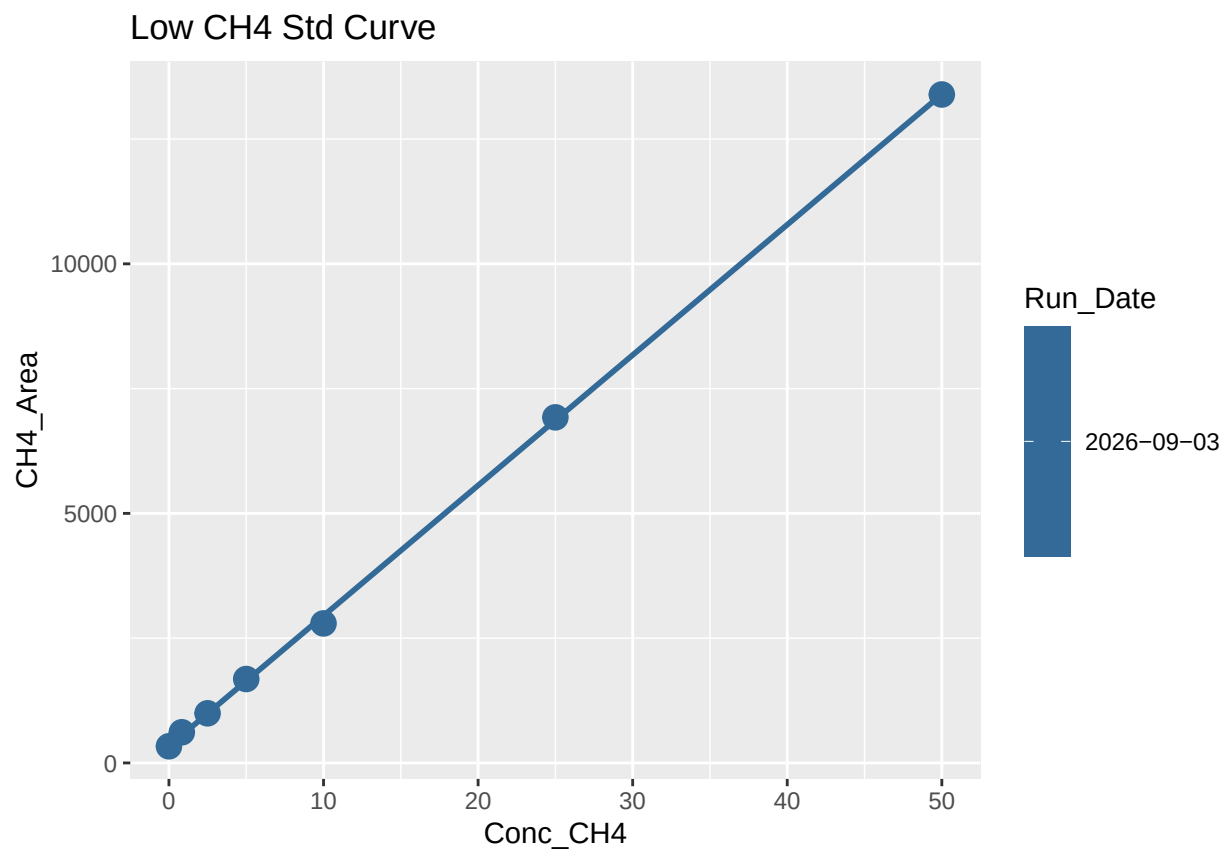
```

1 Assess standard curves CH4

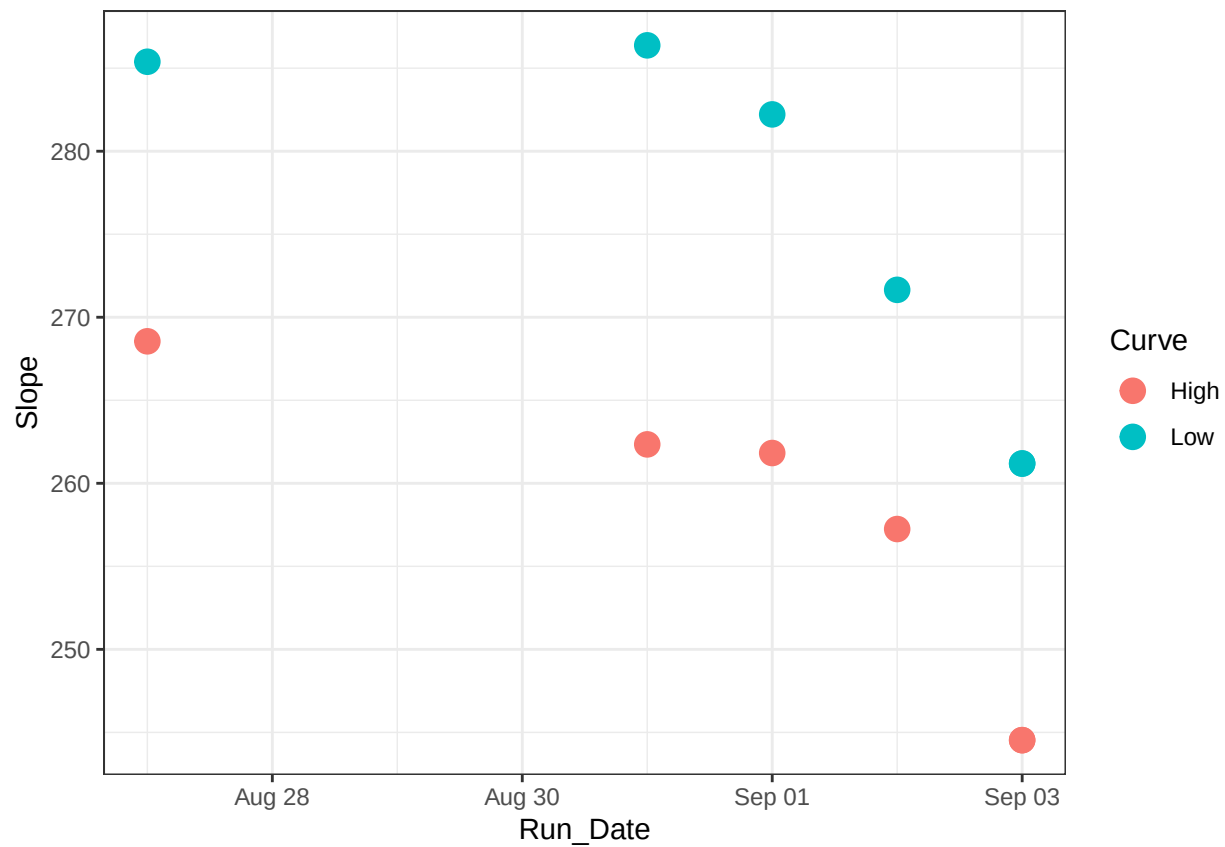
```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```



##	Run_Date	Curve	R2	Slope	Intercept	Analyte
## 1	2026-08-27	High	0.9986559	268.551423	9535.1095	CH4
## 2	2026-08-27	Low	0.9995601	285.387517	115.8345	CH4
## 3	2026-08-27	Full	0.9984309	2.396263	254.4449	CO2
## 4	2026-08-27	High	0.9889439	1014.018375	3704.2202	N2O
## 5	2026-08-27	Low	0.9986226	1454.680571	214.6734	N2O
## 6	2026-08-31	High	0.9987205	262.344419	9865.8585	CH4

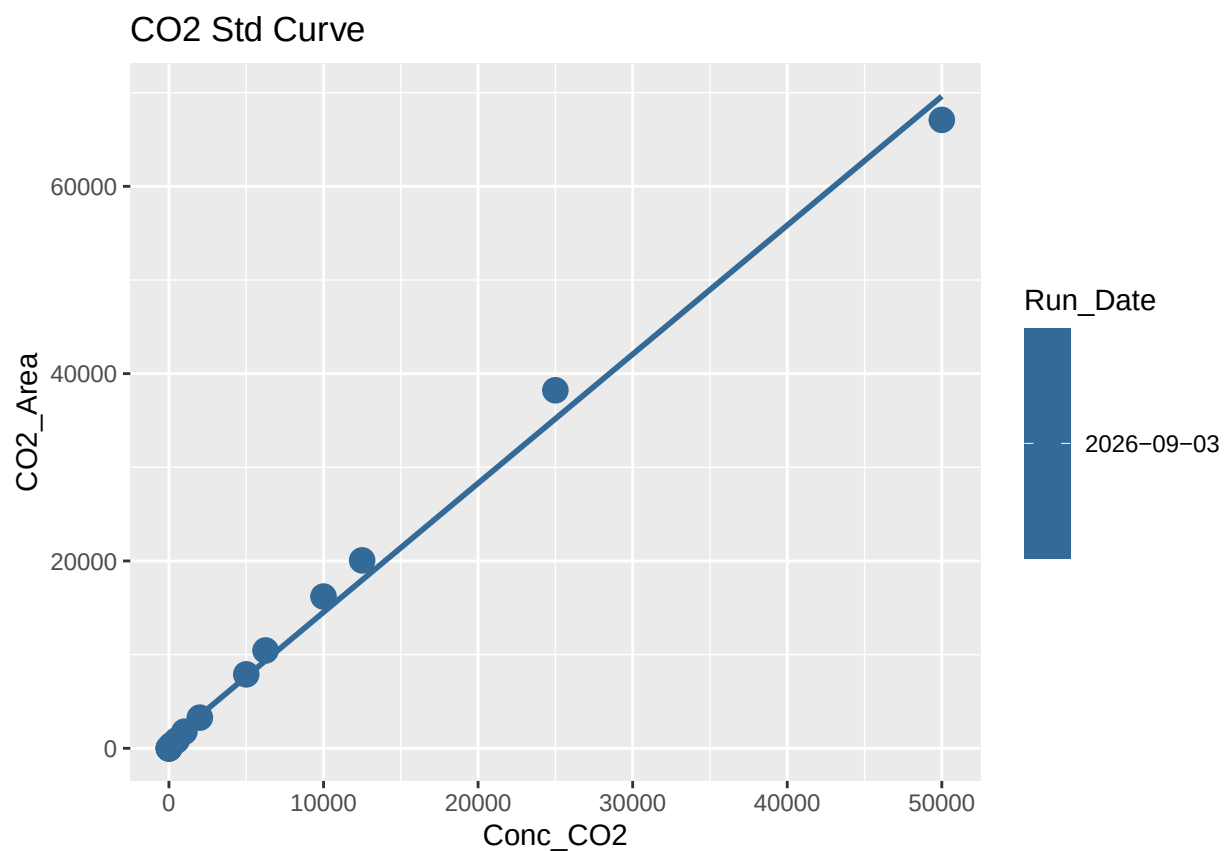


```
## [1] "High_R2 is Good-2026-09-03"
```

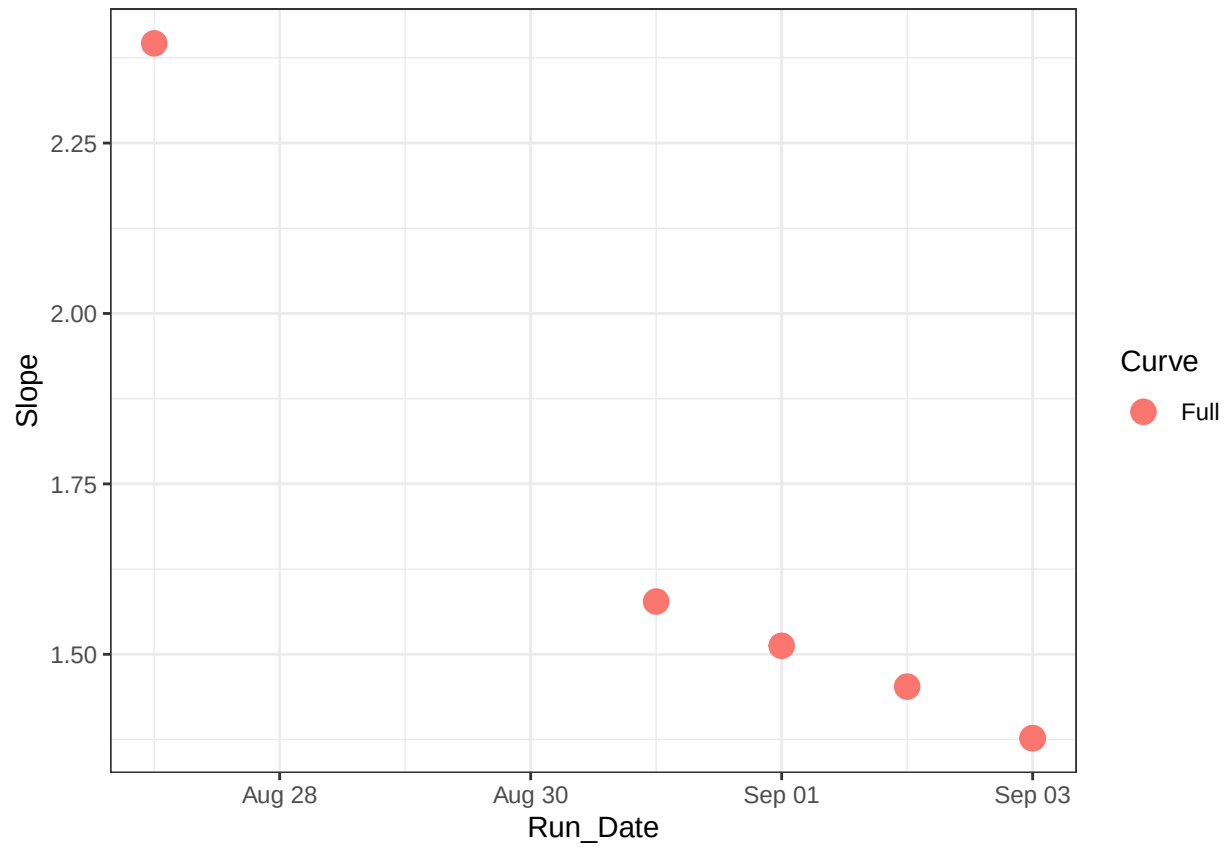
```
## [1] "Low_R2 is Good-2026-09-03"
```

2 Assess standard curves CO2

```
## 'geom_smooth()' using formula = 'y ~ x'
```



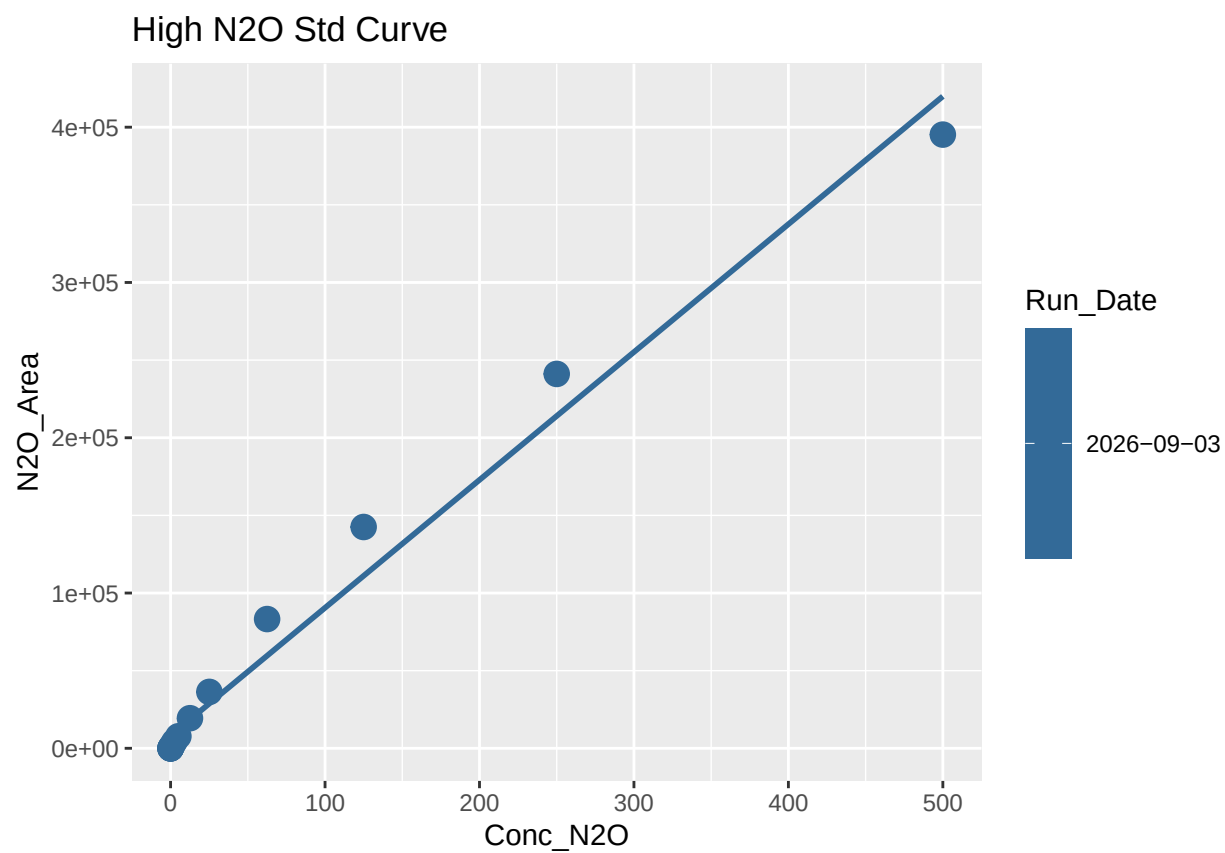
##	Run_Date	Curve	R2	Slope	Intercept	Analyte
## 1	2026-08-27	High	0.9986559	268.551423	9535.1095	CH4
## 2	2026-08-27	Low	0.9995601	285.387517	115.8345	CH4
## 3	2026-08-27	Full	0.9984309	2.396263	254.4449	CO2
## 4	2026-08-27	High	0.9889439	1014.018375	3704.2202	N2O
## 5	2026-08-27	Low	0.9986226	1454.680571	214.6734	N2O
## 6	2026-08-31	High	0.9987205	262.344419	9865.8585	CH4



```
## [1] "C02_R2 is Good-2026-09-03"
```

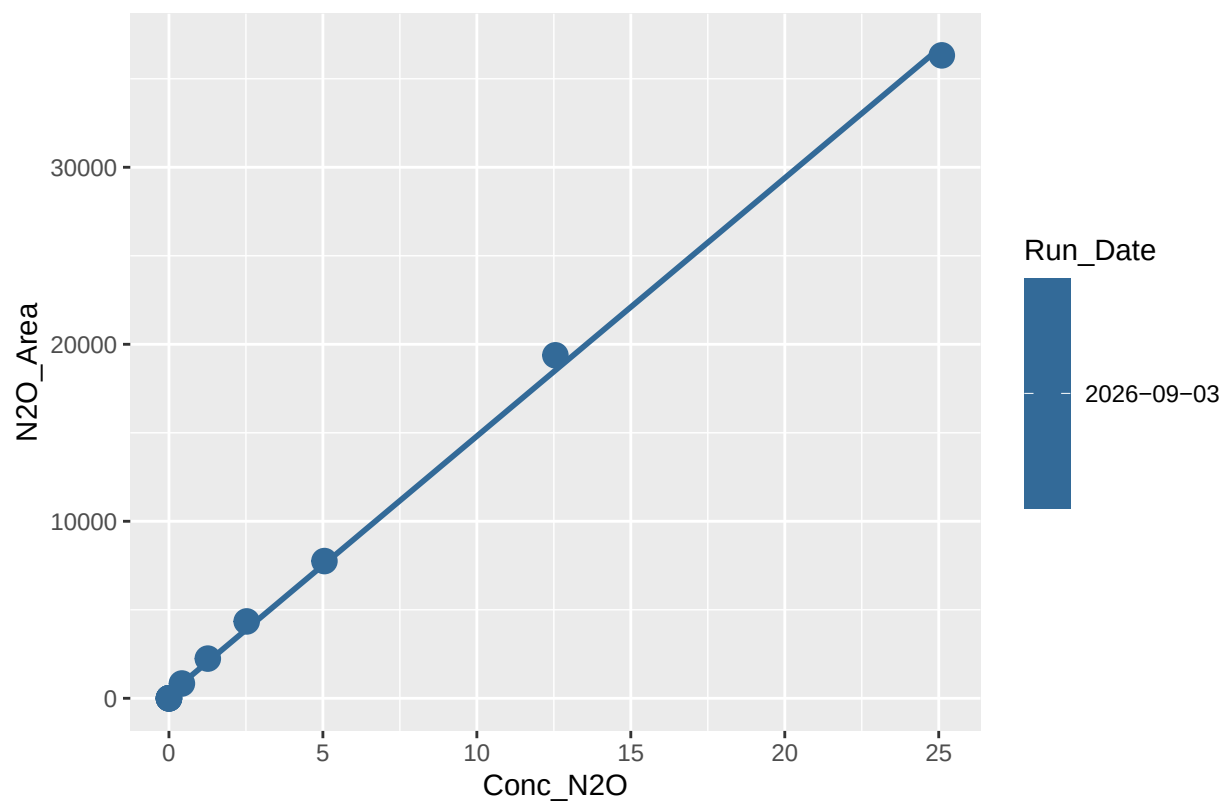
3 Assess standard curves N2O

```
## 'geom_smooth()' using formula = 'y ~ x'
```

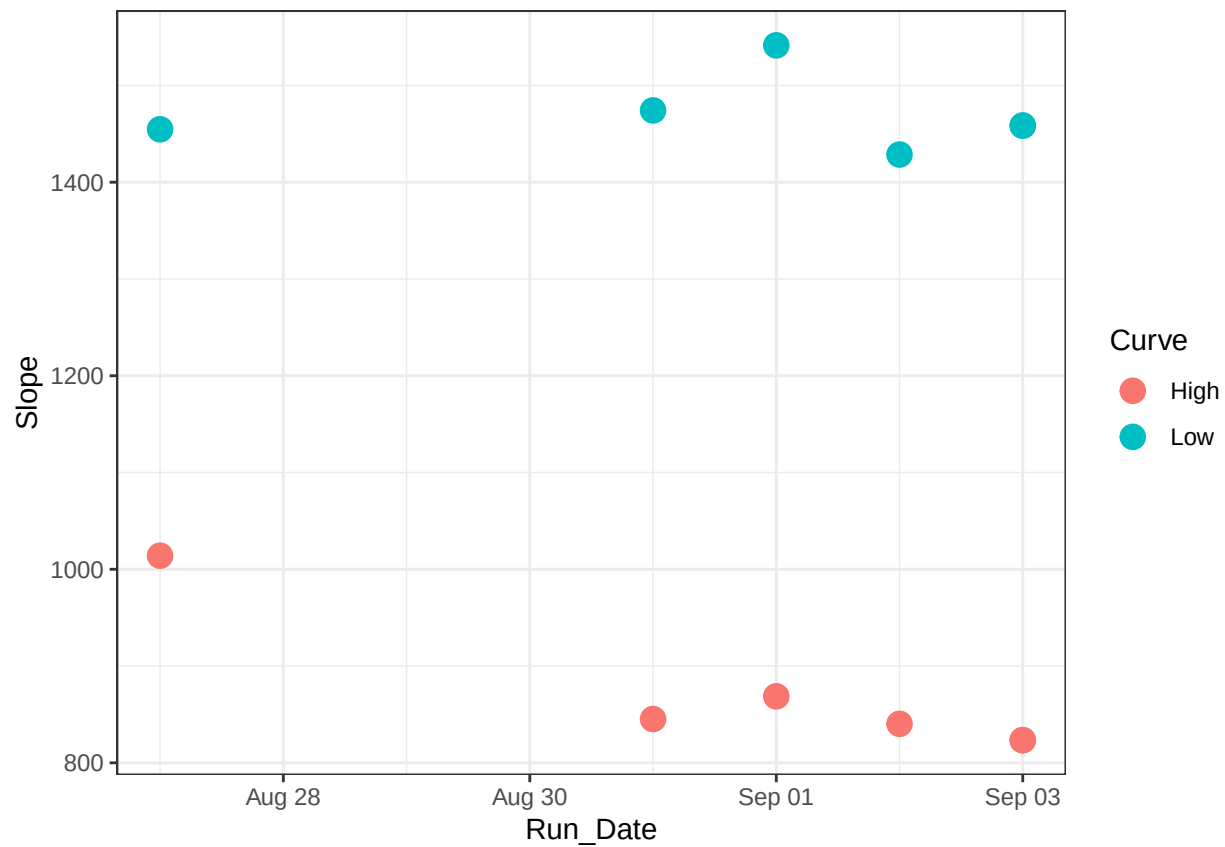


```
## 'geom_smooth()' using formula = 'y ~ x'
```

Low N2O Std Curve



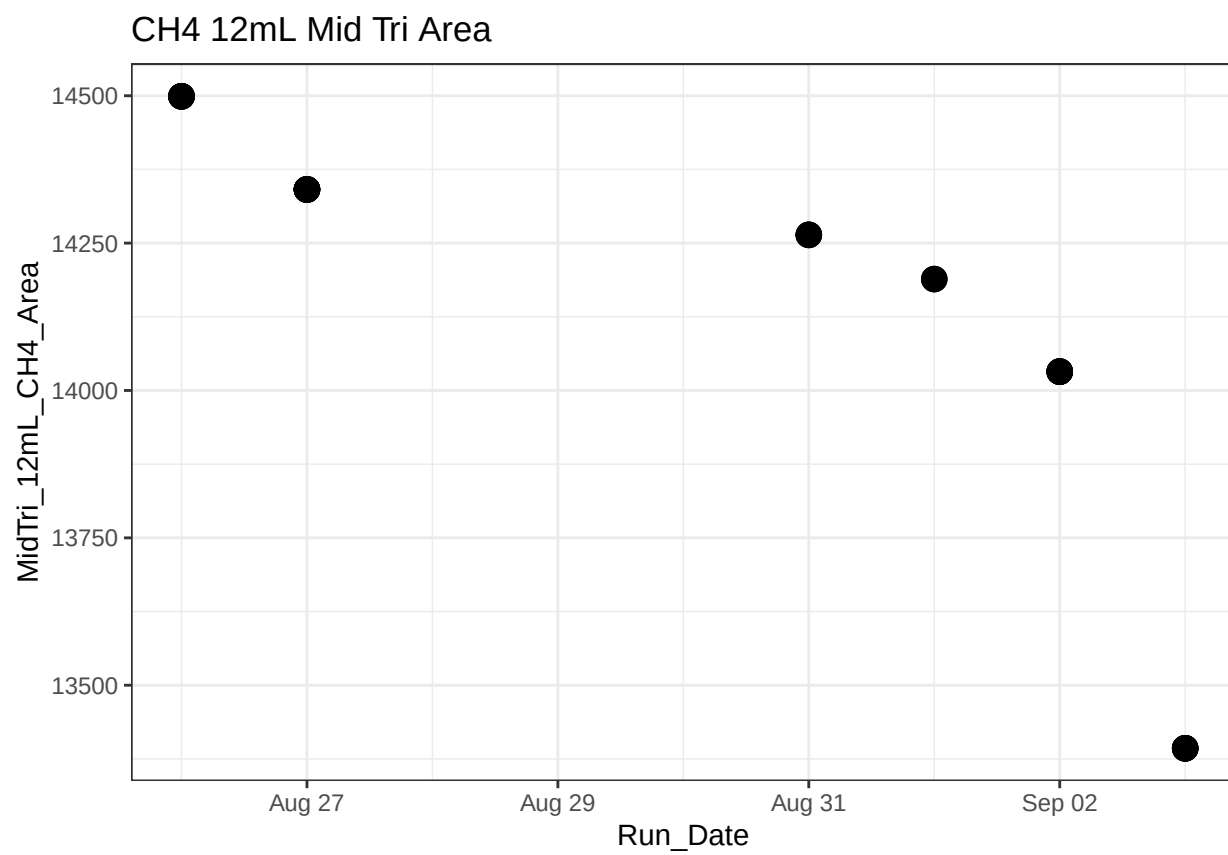
##	Run_Date	Curve	R2	Slope	Intercept	Analyte
## 1	2026-08-27	High	0.9986559	268.551423	9535.1095	CH4
## 2	2026-08-27	Low	0.9995601	285.387517	115.8345	CH4
## 3	2026-08-27	Full	0.9984309	2.396263	254.4449	CO2
## 4	2026-08-27	High	0.9889439	1014.018375	3704.2202	N2O
## 5	2026-08-27	Low	0.9986226	1454.680571	214.6734	N2O
## 6	2026-08-31	High	0.9987205	262.344419	9865.8585	CH4

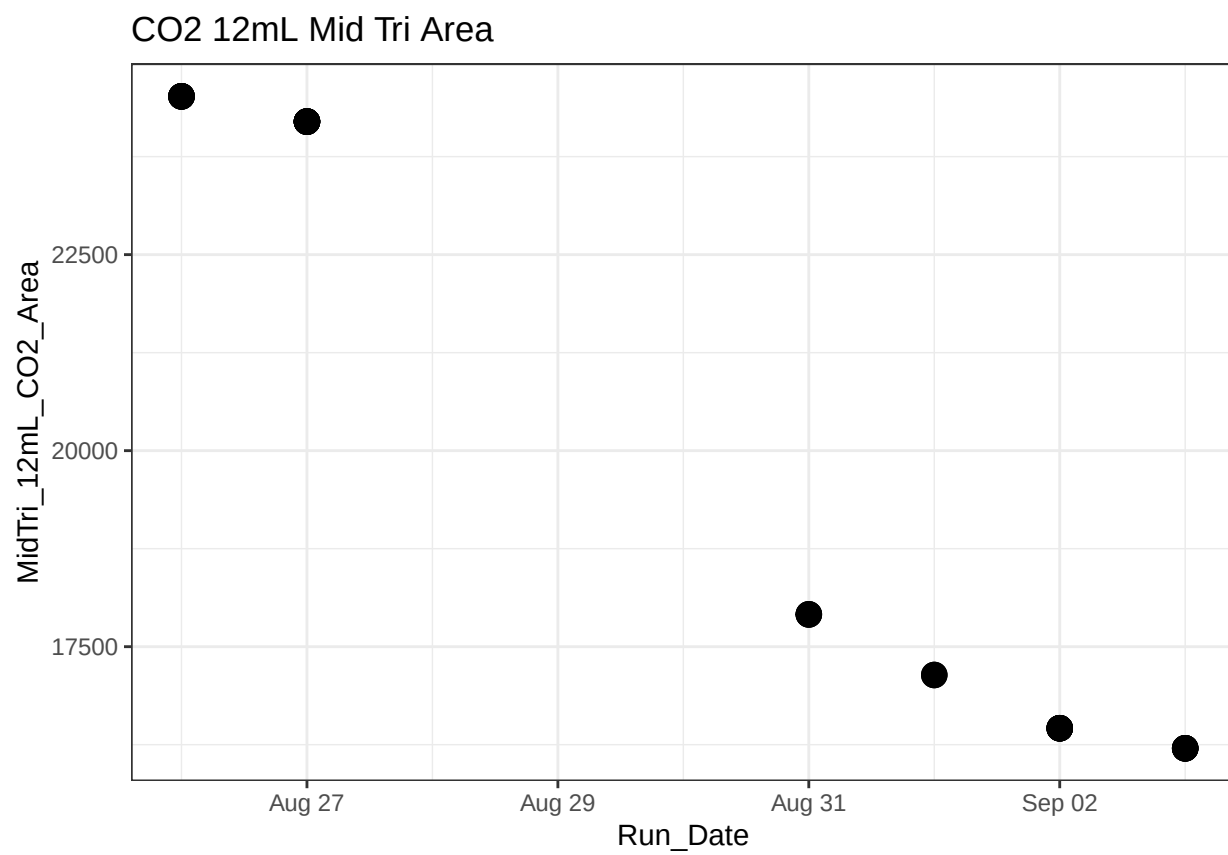


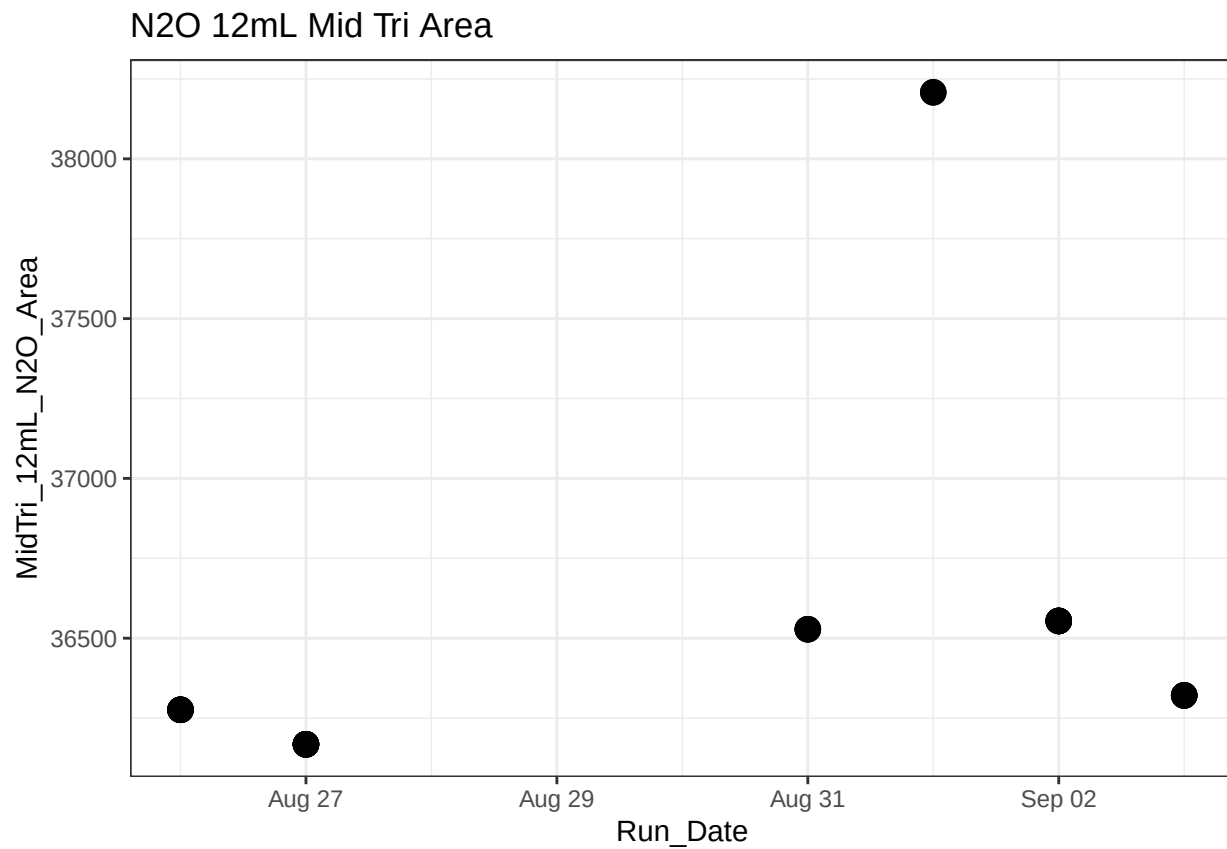
```
## [1] "High_R2 is Bad-2026-09-03"
```

```
## [1] "Low_R2 is Good-2026-09-03"
```

```
#Plot Check Areas over Time
```







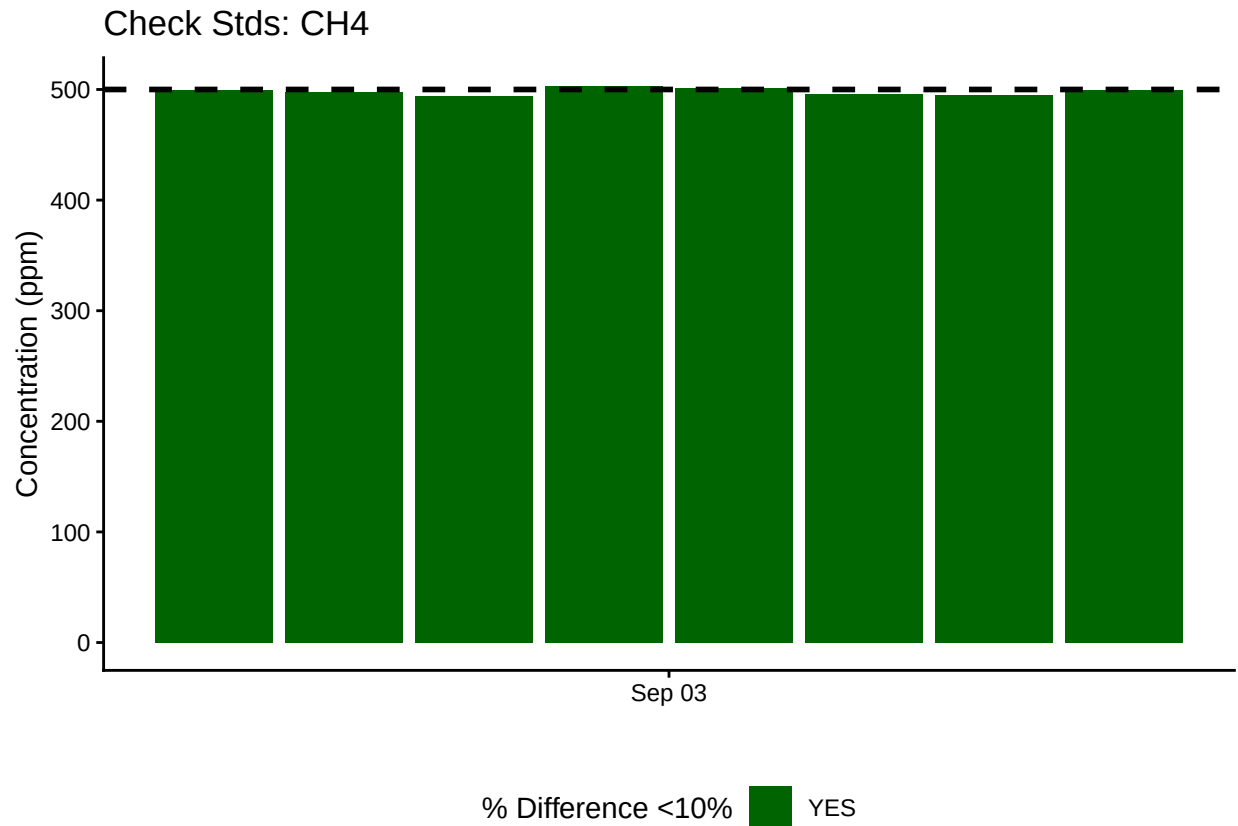
3.1 Now calculate the CH₄ in ppm

3.2 Now calculate the CO₂ in ppm

3.3 Now calculate the N₂O in ppm

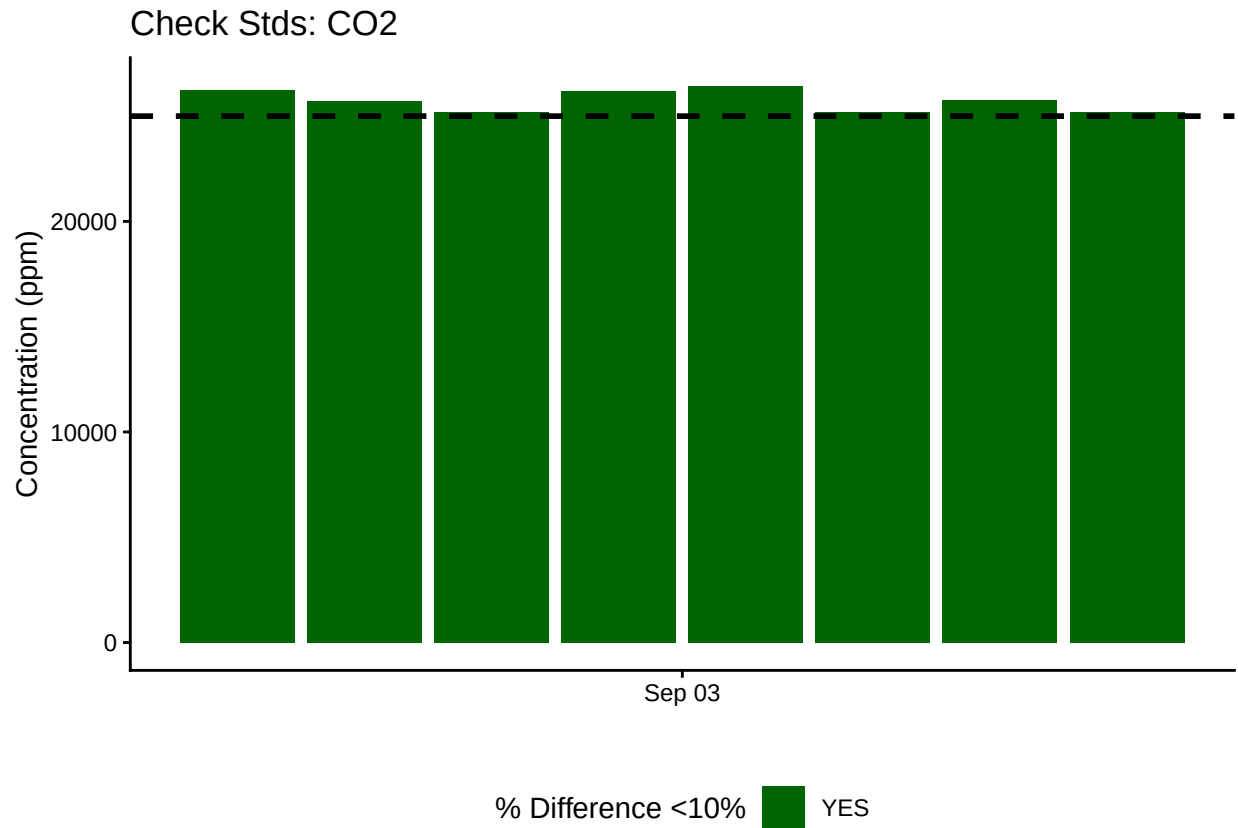
4 Aanlayze the CH₄ Check Standards

>60% of CH₄ Check Standards are within range of expected concentration - PROCEED



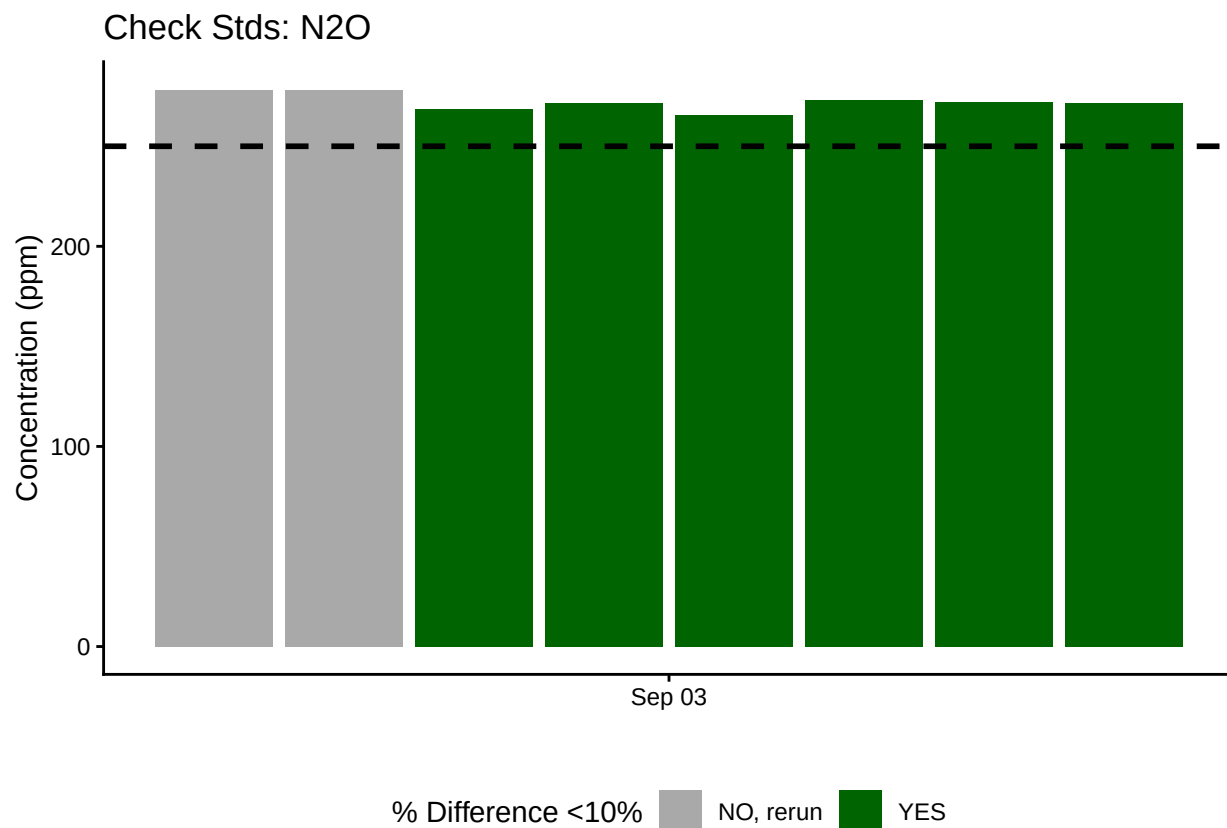
5 Aanlayze the CO2 Check Standards

>60% of CO2 Check Standards are within range of expected concentration - PROCEED



6 Aanlayze the N2O Check Standards

>60% of N2O Check Standards are within range of expected concentration - PROCEED



7 Analyze the CH₄ Blanks

Assess CH₄ Blanks

>60% of CH₄ Blanks are below the lower 25% quartile of samples - PROCEED

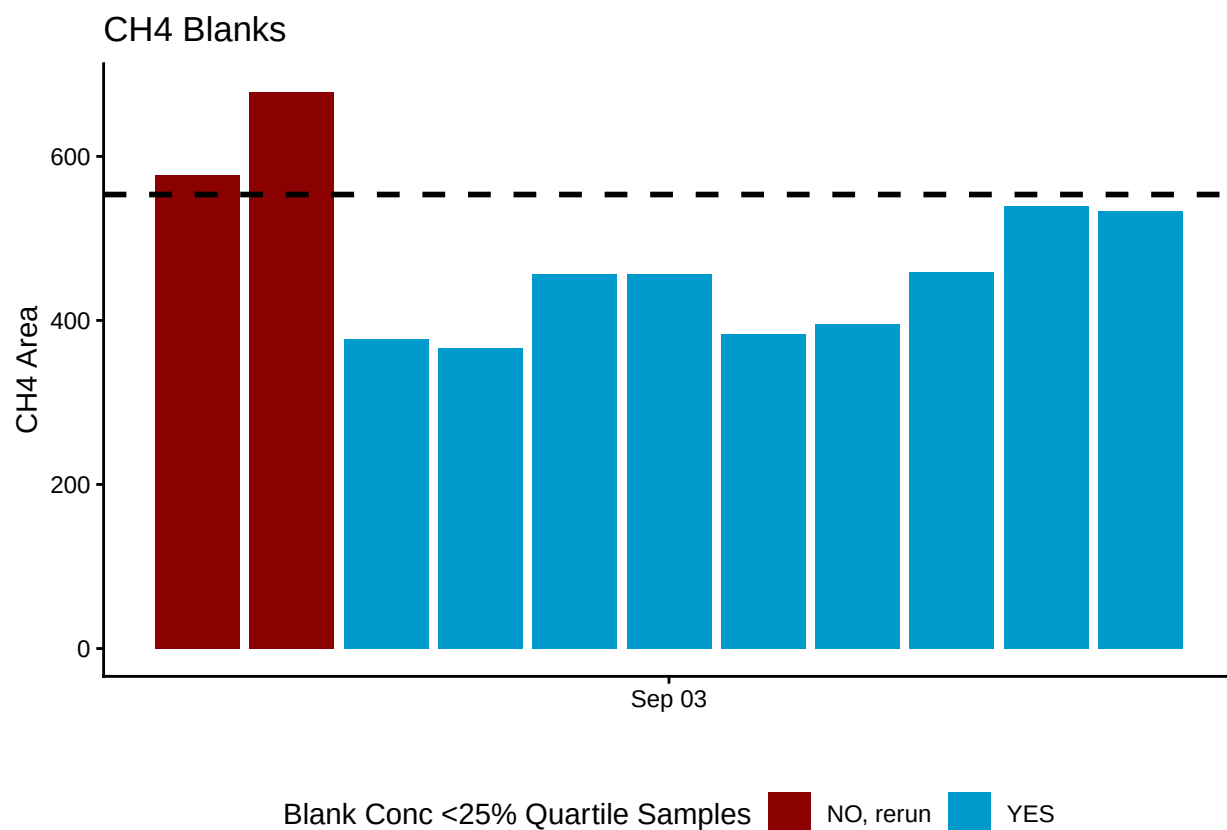


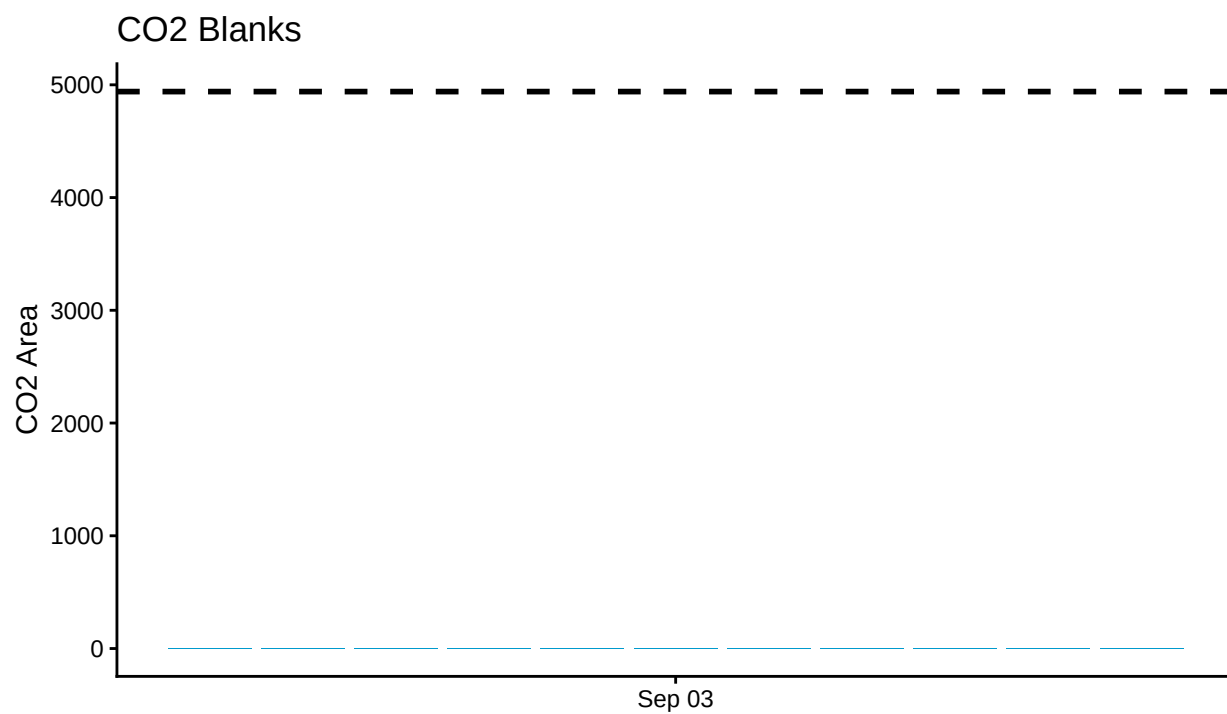
Table 1: Mean Area of Blanks

Test	Blank_Mean_Area
CH4	475

8 Analyze the CO2 Blanks

```
## Assess CO2 Blanks
```

```
## >60% of CO2 Blanks are below the lower 25% quartile of samples - PROCEED
```



Blank Conc <25% Quartile Samples ☒ YES

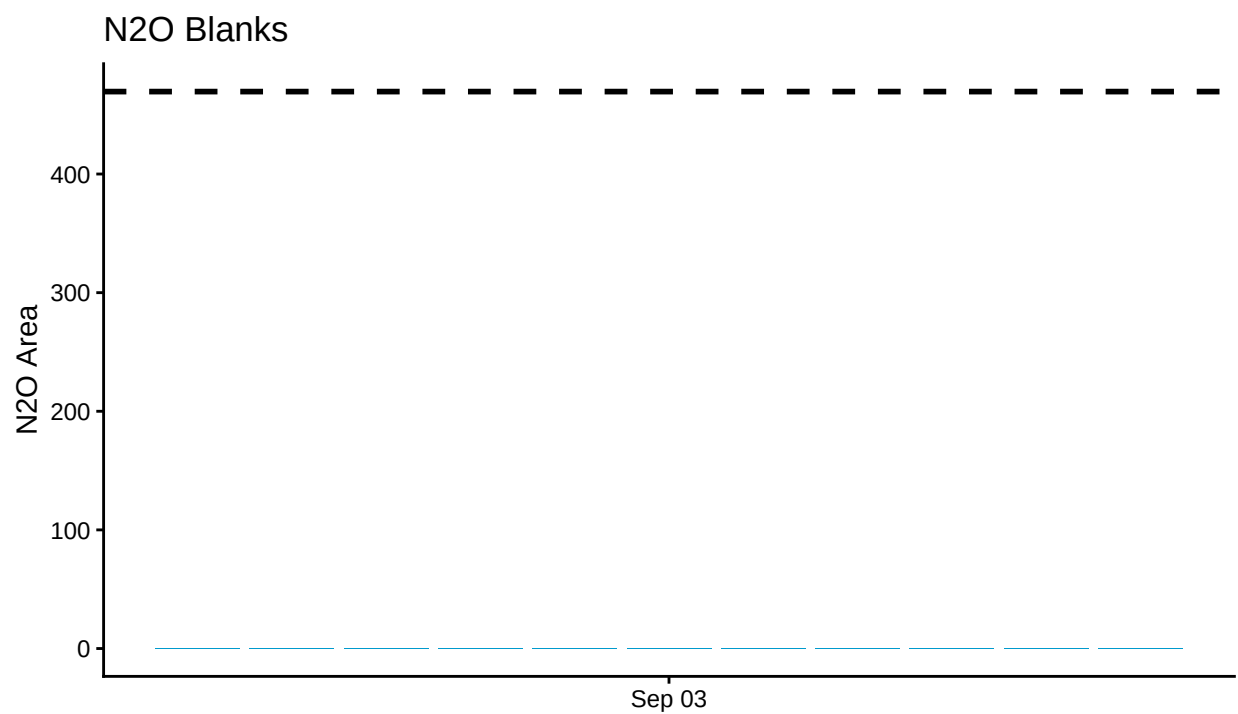
Table 2: Mean Area of Blanks

Test	Blank_Mean_Area
CO2	0

9 Analyze the N2O Blanks

Assess N2O Blanks

>60% of N2O Blanks are below the lower 25% quartile of samples - PROCEED

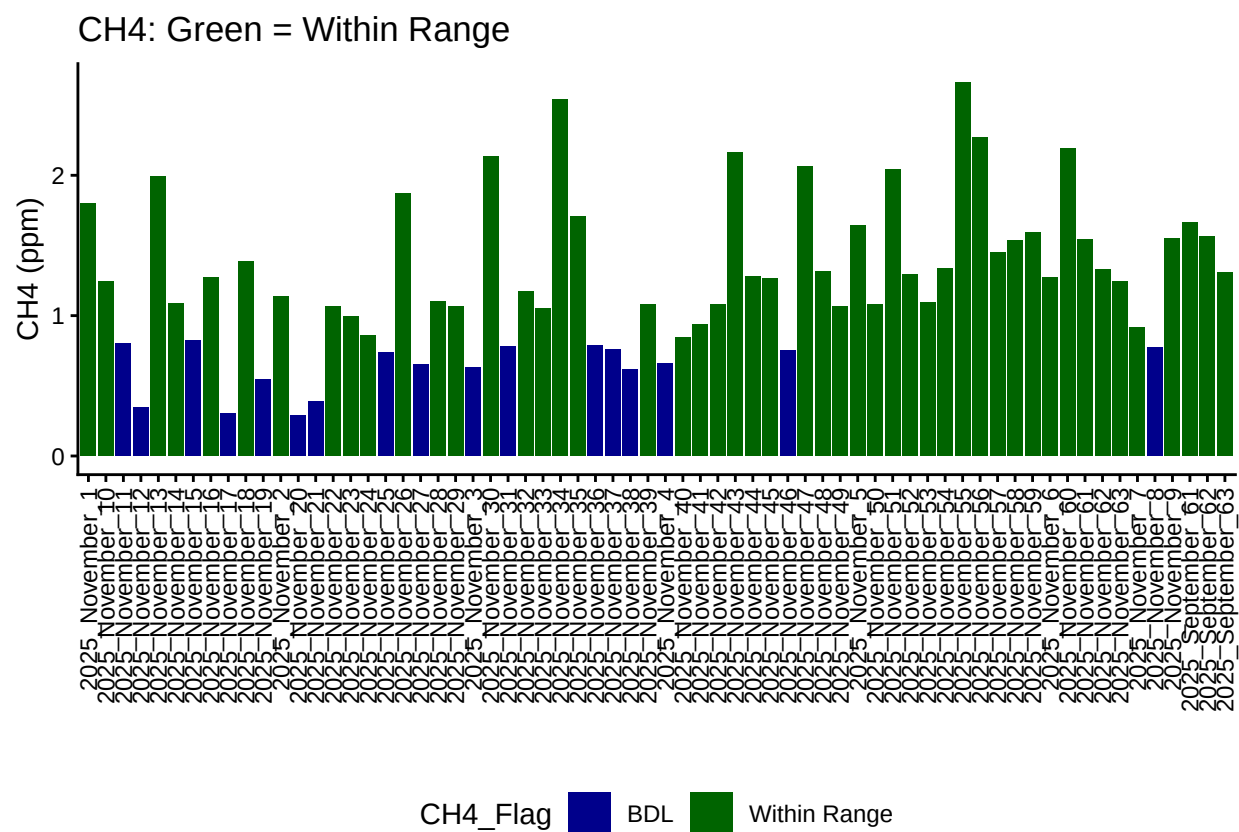


Blank Conc <25% Quartile Samples ■ YES

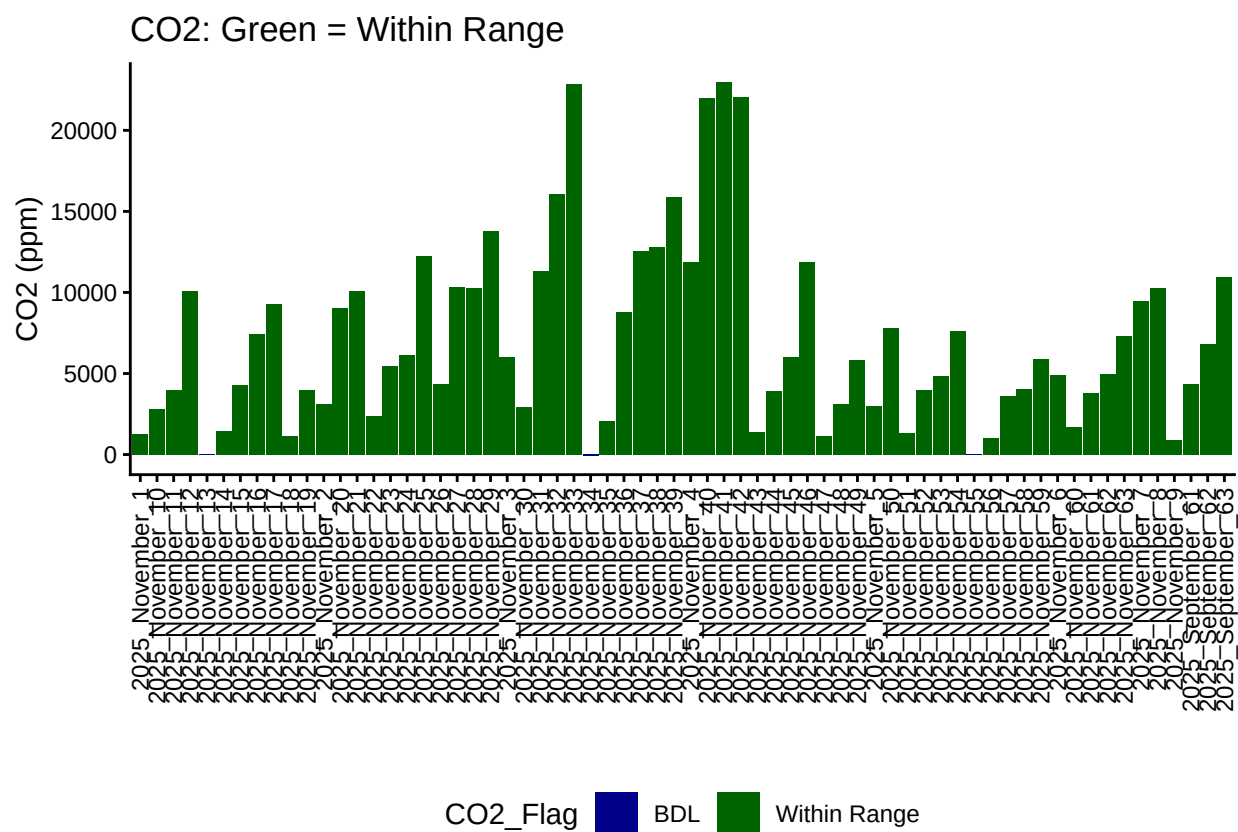
Table 3: Mean Area of Blanks

Test	Blank_Mean_Area
N2O	0

10 Check Range - CH4

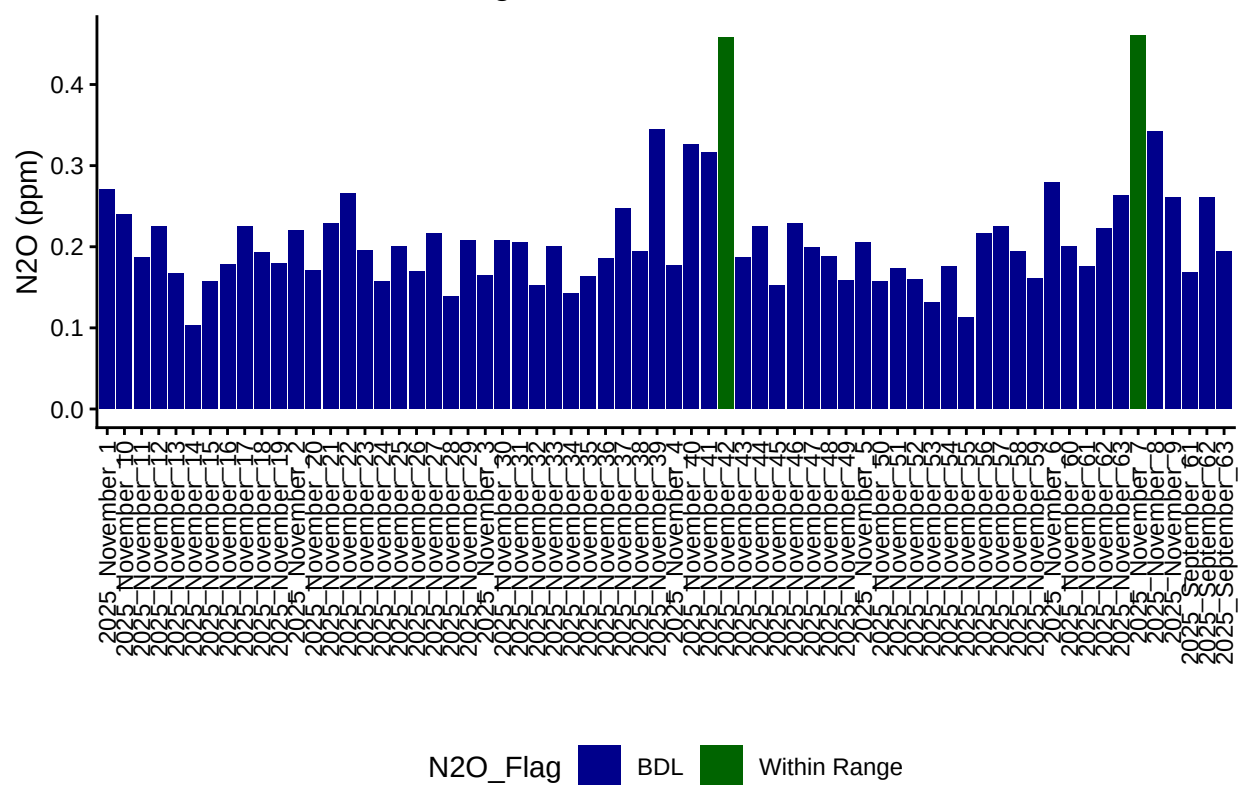


11 Check Range - CO2

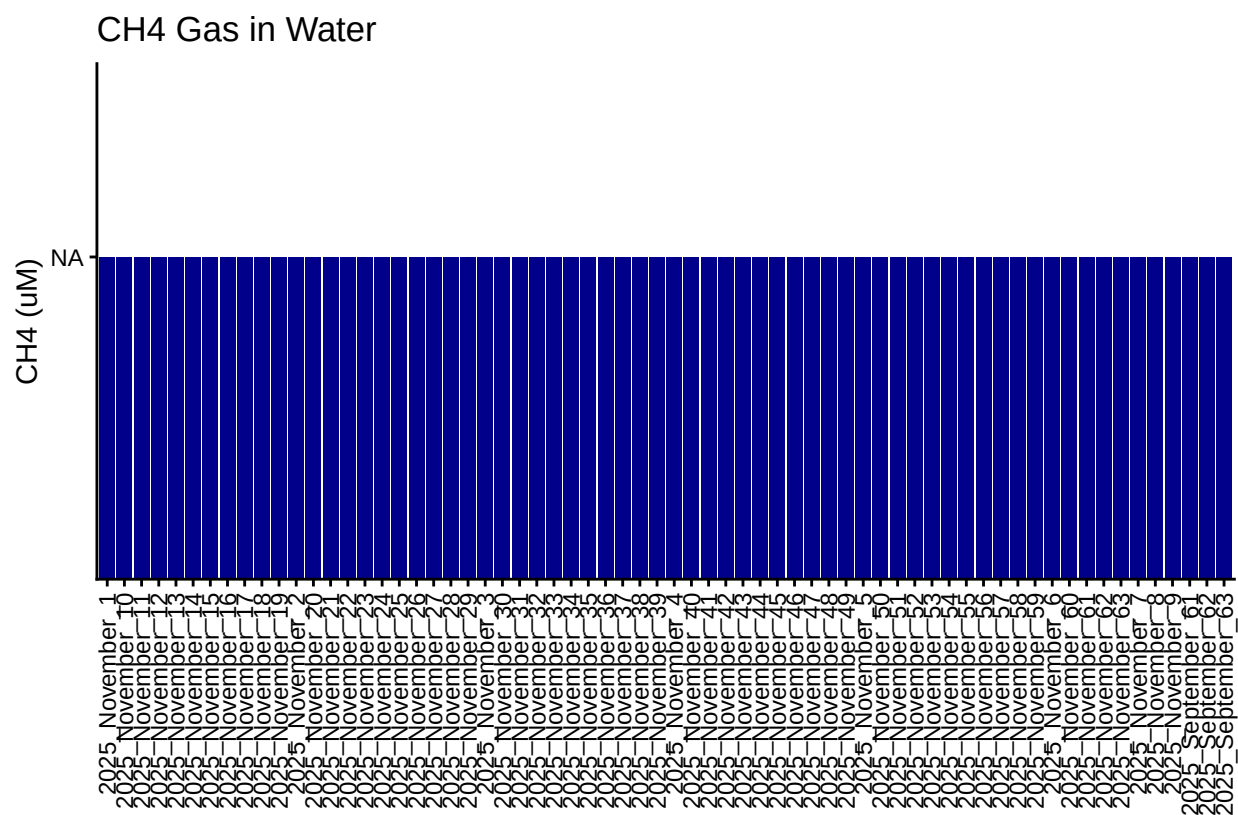


12 Check Range - N2O

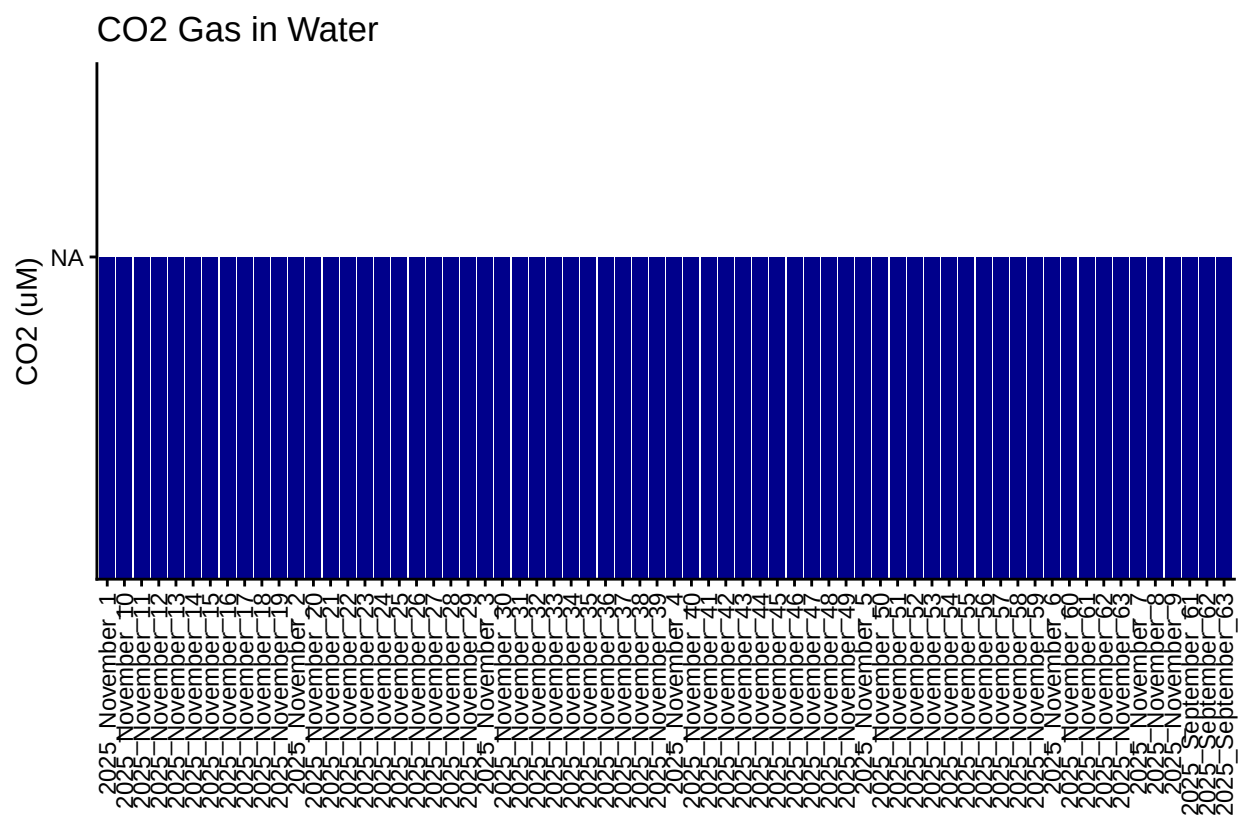
N2O: Green = Within Range



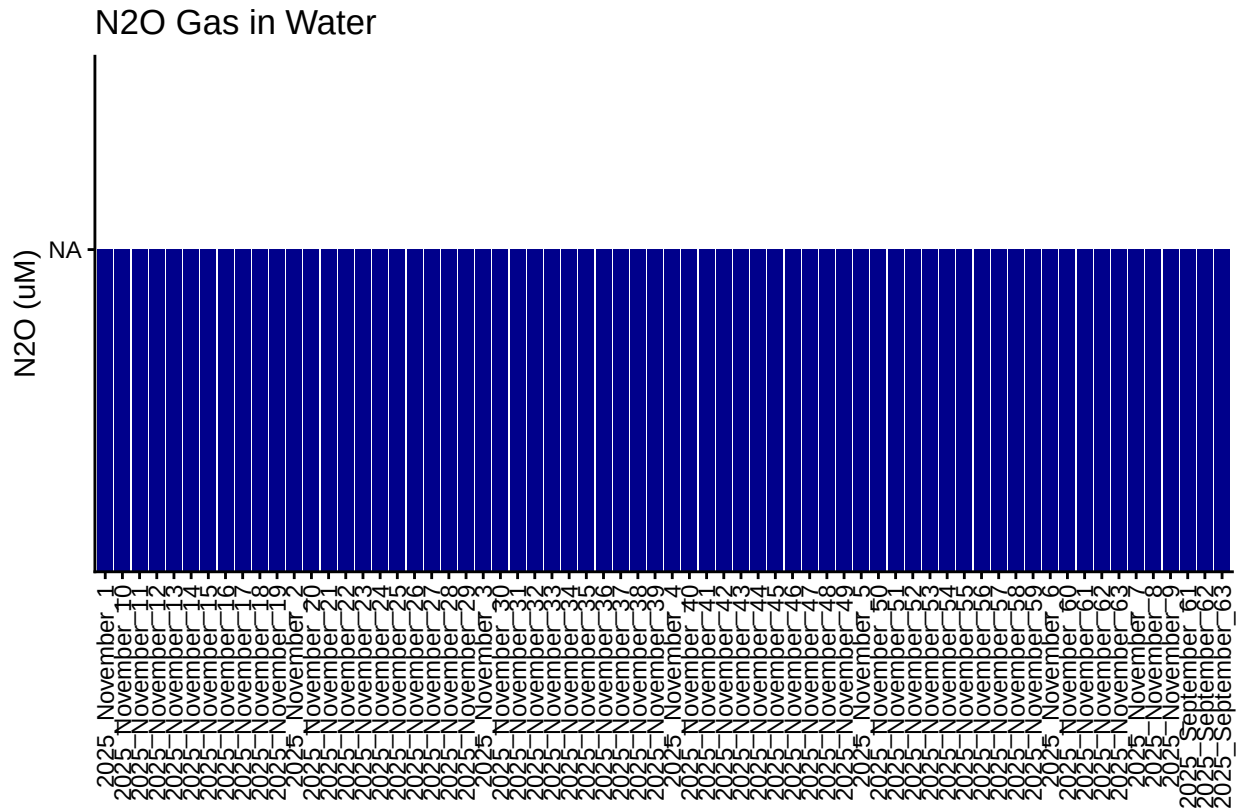
13 Calculate gas in water - CH₄



14 Calculate gas in water - CO2



15 Calculate gas in water - N2O



16 Merge samples with Metadata & check all samples are present

```
## Merge Samples with Metadata
```

```
## All sample IDs are present in data
```

16.1 Organize Data

```
##      Sample_ID  Type  Run_Date CH4_Area Conc_CH4      CH4_RTime_Flag
## 1 2025_November_1 Sample 2026-09-03    806      NA Correct - CH4 RTime
## 2 2025_November_2 Sample 2026-09-03    634      NA Correct - CH4 RTime
## 3 2025_November_3 Sample 2026-09-03    502      NA Correct - CH4 RTime
## 4 2025_November_4 Sample 2026-09-03    508      NA Correct - CH4 RTime
## 5 2025_November_5 Sample 2026-09-03    766      NA Correct - CH4 RTime
## 6 2025_November_6 Sample 2026-09-03    668      NA Correct - CH4 RTime
##      CO2_Area Conc_CO2      CO2_RTime_Flag N2O_Area Conc_N2O      N2O_RTime_Flag
## 1    2526      NA Correct - CO2 RTime    617      NA Correct - N2O RTime
## 2    5029      NA Correct - CO2 RTime    545      NA Correct - N2O RTime
## 3    9073      NA Correct - CO2 RTime    464      NA Correct - N2O RTime
## 4   17074      NA Correct - CO2 RTime    482      NA Correct - N2O RTime
## 5    4916      NA Correct - CO2 RTime    522      NA Correct - N2O RTime
```

```

## 6      7469      NA Correct - CO2 RTime      631      NA Correct - N2O RTime
## FID_Peak_Flag ECD_Peak_Flag CH4_Conc_ppm      CH4_Flag CO2_Conc_ppm
## 1      Correct      Correct      1.7971914 Within Range      1283.567
## 2      Correct      Correct      1.1386711 Within Range      3101.282
## 3      Correct      Correct      0.6332951      BDL      6038.094
## 4      Correct      Correct      0.6562668      BDL      11848.537
## 5      Correct      Correct      1.6440472 Within Range      3019.220
## 6      Correct      Correct      1.2688438 Within Range      4873.246
##      CO2_Flag N2O_Conc_ppm N2O_Flag CH4_Conc_uM CO2_Conc_uM N2O_Conc_uM
## 1 Within Range      0.2700012      BDL      NA      NA      NA
## 2 Within Range      0.2206354      BDL      NA      NA      NA
## 3 Within Range      0.1650989      BDL      NA      NA      NA
## 4 Within Range      0.1774403      BDL      NA      NA      NA
## 5 Within Range      0.2048658      BDL      NA      NA      NA
## 6 Within Range      0.2796001      BDL      NA      NA      NA

## Sample1 Sample_Year Sample_Month Sample_Day Sample_Time Time_Zone G_W
## 1      SGW      2025      November      10      1140      EST      G
## 2      SGW      2025      November      10      1141      EST      G
## 3      SGW      2025      November      10      1142      EST      G
## 4      SGW      2025      November      10      1143      EST      G
## 5      SGW      2025      November      10      1132      EST      G
## 6      SGW      2025      November      10      1133      EST      G
## Sample_Plot Sample_Grid Sample_Depth Field_Notes Lab_Notes CH4_Conc_ppm
## 1      Control      C3      10      NA      NA      1.7971914
## 2      Control      C3      20      NA      NA      1.1386711
## 3      Control      C3      35      NA      NA      0.6332951
## 4      Control      C3      50      NA      NA      0.6562668
## 5      Control      C6      10      NA      NA      1.6440472
## 6      Control      C6      20      NA      NA      1.2688438
## CH4_Conc_uM CH4_Conc_Flag CO2_Conc_ppm CO2_Conc_uM CO2_Conc_Flag N2O_Conc_ppm
## 1      NA      Within Range      1283.567      NA      Within Range      0.2700012
## 2      NA      Within Range      3101.282      NA      Within Range      0.2206354
## 3      NA      BDL      6038.094      NA      Within Range      0.1650989
## 4      NA      BDL      11848.537      NA      Within Range      0.1774403
## 5      NA      Within Range      3019.220      NA      Within Range      0.2048658
## 6      NA      Within Range      4873.246      NA      Within Range      0.2796001
## N2O_Conc_uM N2O_Conc_Flag      QAQC_Flags Analysis_rundate Run_notes
## 1      NA      BDL N2O_R2 is Bad      2026-09-03
## 2      NA      BDL N2O_R2 is Bad      2026-09-03
## 3      NA      BDL N2O_R2 is Bad      2026-09-03
## 4      NA      BDL N2O_R2 is Bad      2026-09-03
## 5      NA      BDL N2O_R2 is Bad      2026-09-03
## 6      NA      BDL N2O_R2 is Bad      2026-09-03

```

16.2 Visualize Data CH4

```
## Visualize Data CH4
```

TEMPEST: Soil Gas Well CH4



16.3 Visualize Data CO2

```
## Visualize Data CO2
```

TEMPEST: Soil Gas Well CO2



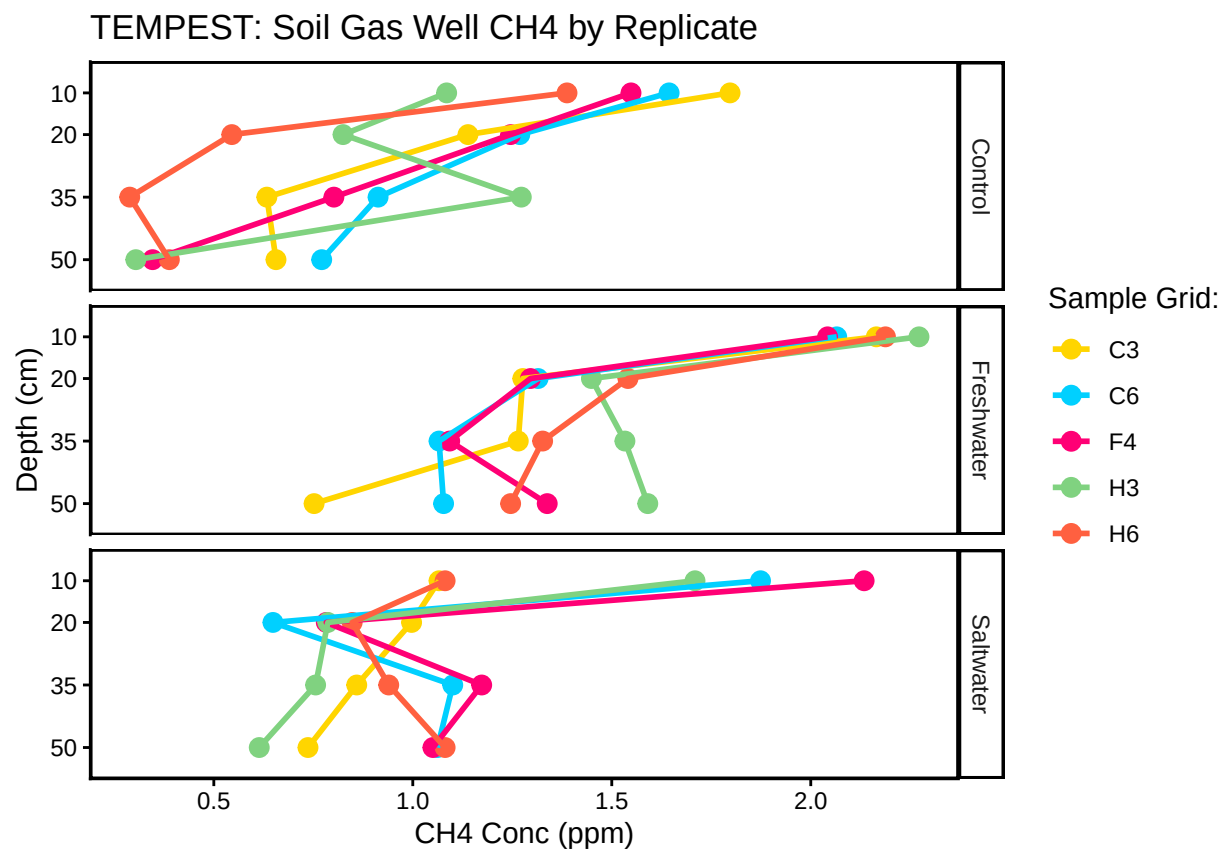
16.4 Visualize Data N2O

```
## Visualize Data N2O
```

TEMPEST: Soil Gas Well N2O

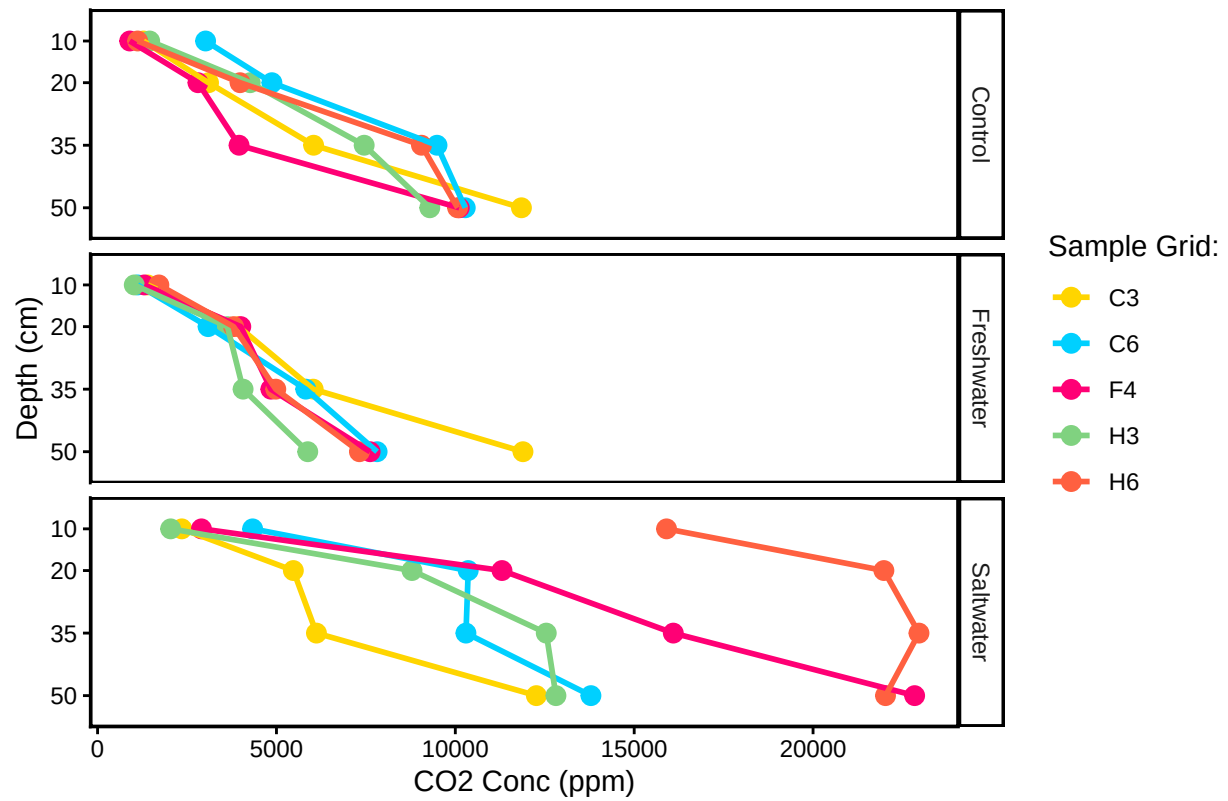


#Visualize Data CH4 by Replicate



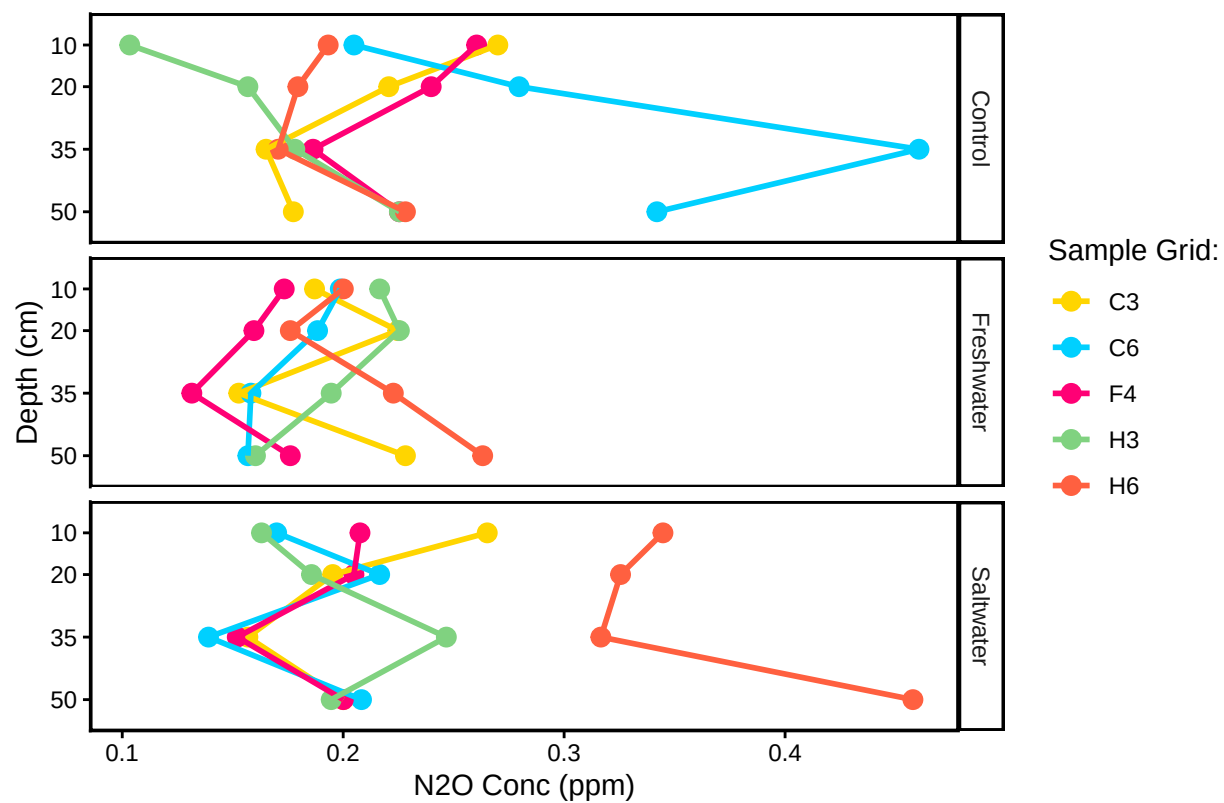
#Visualize Data CO2 by Replicate

TEMPEST: Soil Gas Well CO2 by Replicate



#Visualize Data N2O by Replicate

TEMPEST: Soil Gas Well N2O by Replicate



##Export Data

#end